

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Unburnt fuel in vehicle exhaust gases (1) • Evaporation from fuel tanks / spillage at filling stations (1) • Solvents (paints, cleaning products) (1) • Aerosol propellants (1) • Gas leaks / fossil fuel exploitation (methane) (1)	3
01.2	<i>1 mark for each point.</i> • Vapours displaced from the vehicle tank are collected, condensed and returned to the main storage tank (1) • Vapours are passed through an activated carbon filter where the hydrocarbons adsorb onto the carbon (1)	2
01.3	<i>1 mark for each point.</i> • A platinum / palladium catalyst on a high surface-area honeycomb in the exhaust pipe (1) • Carbon monoxide and hydrocarbons are oxidised to carbon dioxide (and water) (1) • NO_x are chemically reduced to nitrogen and oxygen (1)	3
01.4	<i>1 mark for each point.</i> • Binds to haemoglobin (more strongly than oxygen) so haemoglobin cannot carry oxygen from the lungs to tissues (1) • Brain damage / death if enough haemoglobin is inactivated; low doses worsen chronic heart disease (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 249, 254; AQA spec 3.4.3.2.4-3.4.3.2.5.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> • B (about 40:40:20 sand:silt:clay) (1) • Balanced mix gives good drainage and aeration from the sand fraction (1) • Water retention and nutrient (ion) retention from the clay fraction; easy root penetration and cultivation (1)	3
02.2	<i>1 mark for each point.</i> • Large sand particles leave large pore spaces so water drains rapidly (high permeability / aeration) (1) • Nutrient ions adsorb onto the (charged) surfaces of clay particles but not onto sand (1) • With only 5% clay there is little surface to hold ions so nutrients are leached; low water retention too (1)	3
02.3	<i>1 mark for each point.</i> • C (clay) holds much water so has a high thermal capacity: warms and cools slowly; A warms rapidly after cold weather (1) • Clay particles held together by capillary water so roots find it hard to push between them; penetration is easy in sandy A (1)	2
02.4	<i>1 mark for each point.</i> • Shake a known mass of soil with water in a measuring cylinder and leave to settle; particles settle in order of size (sand first, then silt, then clay; organic matter floats) (1) • Measure the depth / volume of each layer to calculate the percentage of each fraction, then use the soil triangle to name the textural class (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-160, 412; AQA spec 3.2.5.1.	
03.1	<i>1 mark for each point.</i> • The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) (1) • Radon travels up cracks in the rock and through foundations into buildings (1) • It becomes concentrated inside enclosed / poorly ventilated buildings; control by ventilating foundations / sumps (1)	3
03.2	<i>1 mark for the correct answer.</i> • $16 + 1.5 = 17.5\%$ (1)	1
03.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Where they live (geology / radon, altitude) (1) • Occupation (radiographer, nuclear worker, aircrew, miner) (1) • Medical history (number of X-rays / CT scans) (1) • Lifestyle / diet (eg shellfish near discharges); smoking (1)	2
03.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it; risks can be reduced (shielding, dose limits) (1) • Symptoms of chronic low-dose exposure take a long time to develop (1) • The same symptoms (cancers) have other causes so effects are hard to attribute (1) • Accurate data on human exposure / doses is often unavailable (especially after accidents) (1) • People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers) (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	
04.1	<i>1 mark for each point.</i> - The scatter / variability of results around each mean; about 68% of values lie within one SD of the mean (1) - The SDs are large relative to the difference in means (1.9 g) so the ranges overlap greatly (1)	2
04.2	<i>1 mark for each point.</i> - t-test (1) - Comparing two means of measured (continuous) data (1)	2
04.3	<i>1 mark for each point.</i> - There is a 30% probability that the difference occurred by random chance (1) - Above the 0.05 (5%) level so the difference is not significant; the null hypothesis (no difference) is accepted (1)	2
04.4	<i>1 mark for each point.</i> - Two correlated variables may both be caused by a third variable that was not investigated (1) - Significance only shows the result is unlikely to be chance; a mechanism / controlled experiment is needed for causation (1)	2
04.5	<i>1 mark for each point.</i> - Increase the number of replicates (reduces the effect of variability) (1) - Standardise other variables (soil, water, light) more closely to reduce scatter (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
05.1	<i>1 mark for each point.</i> - Cost increases as the percentage prevented increases (1) - The relationship is not linear: cost rises slowly at first then very steeply / exponentially above about 80-90% (1)	2
05.2	<i>1 mark for the correct answer.</i> $16.0 - 5.0 = 11.0$ (£ million) (1)	1
05.3	<i>1 mark for each point.</i> - ALARA: emissions should be As Low As Reasonably Achievable (1) - It is rarely practical / affordable to reduce emissions to zero; low-level emissions may have acceptable impacts so an acceptable emission level is set (1) - BATNEEC: Best Available Technology Not Entailing Excessive Cost - choosing the most effective control equipment whose cost is not excessive (1) - The factory would install equipment that gives a high level of control (eg 80-90%) rather than paying £16 million for 99% (1)	4

Question	Marking guidance	Marks
05.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Industry / city may have expanded so there are more pollutant sources and total emissions have risen (1) • Better understanding of the health / environmental effects so stricter standards are set (1) • Improved technology becomes available (BATNEEC changes) / cost of better control falls (1) • Pollutant may have accumulated (persistent) so background levels are higher (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 244-245; AQA spec 3.4.3.1.	
06.1	<i>Correct answer only.</i> • C (1)	1
06.2	<i>1 mark for each point.</i> • Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms (1) • Tropospheric ozone is a secondary pollutant (from NO_x photochemistry) that is toxic to plants and humans and is a greenhouse gas (1)	2
06.3	<i>1 mark for each point.</i> • HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere (1) • So most do not persist long enough to reach the stratosphere where UV would release their chlorine (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
07.1	<i>1 mark for each point.</i> • No large dam / reservoir needed so no flooding of farmland, homes or habitats and no barrier across the whole river (run-of-river / micro-hydro diverts part of the flow) (1) • Suitable for rivers with a small drop in height and lower flow rates that are common in lowland rural areas (1) • Small scale matches local demand; electricity used locally without long transmission lines / grid connection (1) • Low construction cost and simple technology maintained by the community (1) • Helical turbines tolerate high turbidity / debris without damage and fish pass through unharmed; Kaplan turbines with adjustable blades cope with variable flow (1)	5
07.2	<i>1 mark.</i> • Lack of capital / finance, technical knowledge or grid access; seasonal drying of rivers; conflicts with irrigation water; planning / land ownership (1)	1
07.3	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • Output is proportional to flow x efficiency: $100 \times 0.10 \times (40 / 90)$ (1) • = 4.4 kW (accept 4.4-4.5) (1)	2

Question	Marking guidance	Marks
07.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Constant regulated flow removes seasonal extremes: dry-season sandbanks used by nesting birds / turtles lost; high flows that clear gravel spawning beds lost; floodplain no longer flooded (1) • Sudden flow increases for peak generation raise turbidity, wash away species, reduce light for photosynthesis (1) • Sediment trapped so less nutrient replenishment downstream; barrier to migrating fish (salmon, sturgeon) and isolation of populations (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 203-205; AQA spec 3.3.4.1.	
08.1	<i>1 mark for each point.</i> • Most energy comes from non-renewable fossil fuels whose reserves are depleting; the rate of new discoveries has fallen sharply since the 1980s (1) • Use causes unacceptable environmental damage: atmospheric pollution, climate change, oil pollution, habitat loss (1) • Economic / social: depleting resources become more expensive, reducing future affluence; risk of an energy gap if new technologies are not developed in time (1)	3
08.2	<i>1 mark for each point.</i> • Coal mining: habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise (1) • Oil and gas extraction: spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO₂ / CO₂ / smoke (1) • Pipelines causing habitat damage during construction; tanker transport risks (1) • Fuel processing (refineries, coal washing) pollution; embodied energy of equipment and site development for all resources (1)	4
08.3	<i>1 mark for each point.</i> • New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) (1) • Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise (1) • By then depletion would reduce supply faster than replacements could be built, causing shortages (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
09.1	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up (1) • High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) (1) • Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) (1) • Birds of prey are at the top of the food chain so receive the highest concentrations (1) • Sub-lethal doses caused thin eggshells that broke / reduced breeding success; higher doses caused sterility or death (1)	4

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Advantage: low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly (1) • Disadvantage: high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase); chronic effects (memory loss, depression); suspected carcinogens (1)	2
09.3	<i>1 mark.</i> • Pyrethroids are highly toxic to fish (even though mammal toxicity is low) (1)	1
09.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) (1) • They are moderately persistent so remain in the soil long enough to be absorbed (1) • Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats (1) • So bees / pollinators are less exposed than when the crop is sprayed (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Carbon: combustion of fossil fuels and biomass moving carbon from long-term stores to the atmosphere; deforestation reducing photosynthesis; ploughing / drainage increasing aerobic decomposition; new anaerobic environments (paddies, landfill, reservoirs, livestock) releasing methane; marine pollution reducing phytoplankton; consequences: rising CO₂ and CH₄, climate change, ocean acidification, reduced soil organic matter • Nitrogen: Haber process doubling fixation; legume cultivation; drainage increasing nitrification and reducing denitrification; combustion NO_x; sewage and fertiliser leaching; consequences: eutrophication, smogs, acid rain, N₂O warming, NO_x toxicity • Counter-strategies carbon: renewables / nuclear replacing fossil fuels, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, increasing soil organic matter (low tillage, mulching), CCS (pre / post combustion, storage) • Counter-strategies nitrogen: reduced combustion, catalytic converters / urea sprays, biological fixation instead of Haber, legumes and crop rotation, management of biological wastes as fertiliser, low-solubility fertilisers, timing, buffer strips, low tillage, nitrate vulnerable zones • Evaluation: some strategies well established and cheap (rotation, buffer strips), others expensive or unproven (CCS); scale of fossil fuel dependence; feedback and time lags; link to circular economy and low-temperature natural processes; judgement	9
	Model answer - Level 3 (8 marks)	
	Humans have shifted the carbon cycle by burning fossil fuels and biomass, moving carbon from long-term stores into the atmosphere faster than photosynthesis and ocean uptake remove it, while deforestation reduces photosynthesis, ploughing and drainage speed aerobic decomposition, and paddies, landfills and livestock create new anaerobic sources of methane. The nitrogen cycle has changed as much: the Haber process has roughly doubled fixation, legume crops add more, drainage increases nitrification and reduces denitrification, combustion produces NO_x, and leached fertiliser and sewage nitrate cause eutrophication, while nitrous oxide adds to warming.	

Question	Marking guidance	Marks
	Carbon can be counteracted by replacing fossil fuels with renewables and nuclear power, energy conservation, afforestation matched to deforestation and timber use, conserving peat bogs and forests, and increasing soil organic matter through low tillage and mulching, with carbon capture and storage as an unproven option. For nitrogen, catalytic converters and urea sprays reduce combustion NO _x , biological fixation by legumes and rotation can replace manufactured nitrate, biological wastes can be managed as fertiliser, and low-solubility fertilisers, careful timing and buffer strips reduce leaching.	
	Some strategies are cheap and established - rotation, buffer strips, fertiliser timing - but others are expensive or untested, and the scale of fossil-fuel dependence, together with feedback and time lags, means the atmosphere will respond slowly. The strategies that mimic natural low-temperature cycling, as in the circular economy, are the most sustainable; the most effective is simply to leave carbon in the ground.	
	Weaker answer - Level 1 (3 marks)	
	Humans have changed the carbon cycle by burning fossil fuels which releases carbon dioxide, and by cutting down trees. They have changed the nitrogen cycle by using fertilisers which run off into rivers and cause eutrophication. To counteract this we could use renewable energy instead of fossil fuels and plant more trees. We could use less fertiliser. These would help the cycles go back to normal.	
	What makes the difference	
	• Level 1 names one or two changes per cycle; Level 3 covers several processes for each (decomposition, methane sources, Haber process, nitrification, NO_x).	
	• Level 3 connects each counter-strategy to the process it reverses; Level 1's strategies are general.	
	• The evaluation in Level 3 distinguishes established from unproven strategies and comments on scale and time lags - Level 1 has no evaluation.	
10.2	<i>1 mark.</i> • The average time an atom / molecule remains in the reservoir before being transferred out (volume / transfer rate) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 148-157; AQA spec 3.2.4.	

Question	Marking guidance	Marks
11.1	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25